Cambridge (CIE) IGCSE Physics
Paper 2: Multiple Choice (Set B)
Extended

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

/ 40

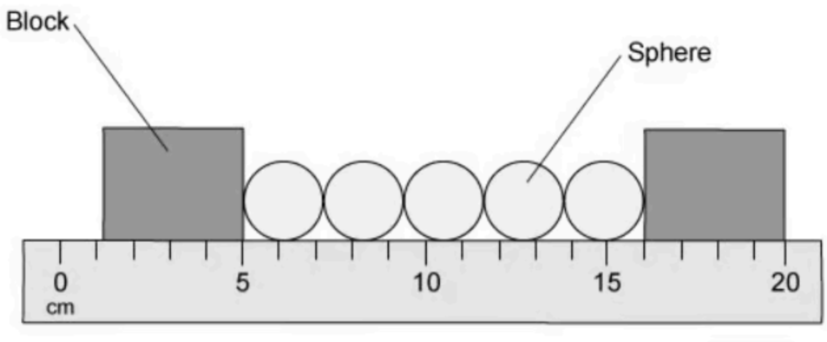
Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/ s^2).

Scan here to mark your mock exam

or visit the mock exams landing page for this course

1 Five identical spheres are placed between two blocks in order to measure their diameter.



What is the diameter of a single sphere?

A. 2.2 cm

B. 2.0 cm

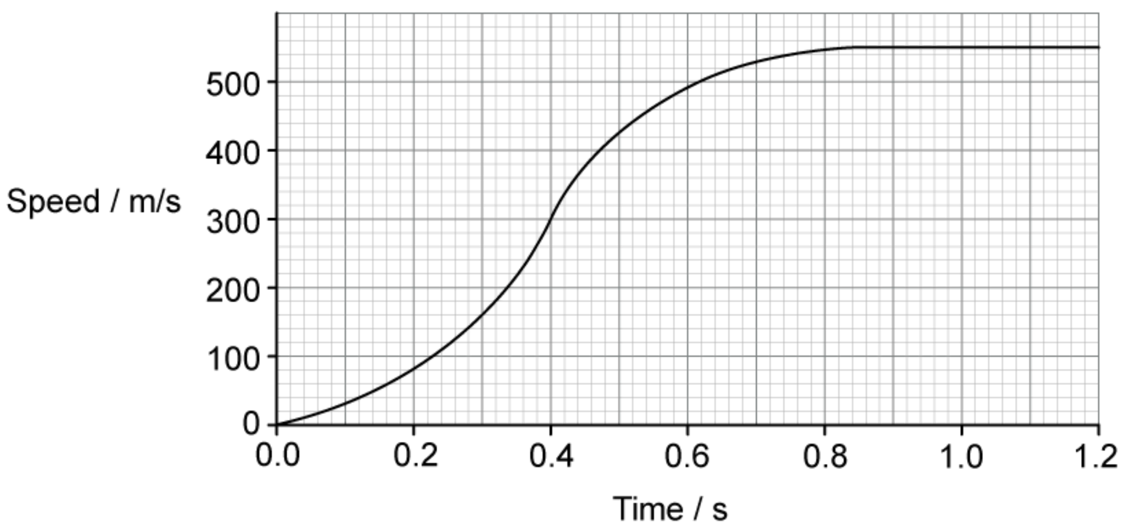
C. 2.1 cm

D. 2.3 cm

(1 mark)

2 A charged particle is accelerated when passing through an electric field.

What is the acceleration of the particle at 0.4 s?



A. 2.0×10^3 m/s^2

B. 2.5 m/s^2

C. 2.5×10^{-3} m/s^2

D. 2.0×10^{-3} m/s^2

(1 mark)

3 A cannonball is dropped from a three story building.

Which row of the table correctly describes both the speed and the acceleration of the cannonball as it falls?

You can ignore air resistance for this question.

	Speed	**Acceleration**
A	constant	constant
B	increasing	constant
C	increasing	increasing
D	constant	increasing

(1 mark)

4 A car is driving around a circular track at a constant speed.

Which statement describes the motion of the car?

A. The car is accelerating because its speed is changing

B. The car is not accelerating as it is moving at a constant speed

C. The car is accelerating because its velocity is changing

D. The car is not accelerating, but its velocity is changing.

(1 mark)

5 Some unknown liquid in a beaker has a mass of 200 g and a volume of 230 cm^3.

The density of water is 1.0 g/cm^3

How does the density of the unknown liquid compare with the density of water?

A. Its density is greater than the density of water

B. Its density is less than the density of water

C. Its density is the same as the density of water

D. With the information given, it is impossible to tell.

(1 mark)

6 After a car crash the car driver's airbag inflates. The airbag then deflates when it is hit by the driver's head.

How does an airbag reduce the risk of injury?

A. Collision time increases, which increases the rate of change of momentum.

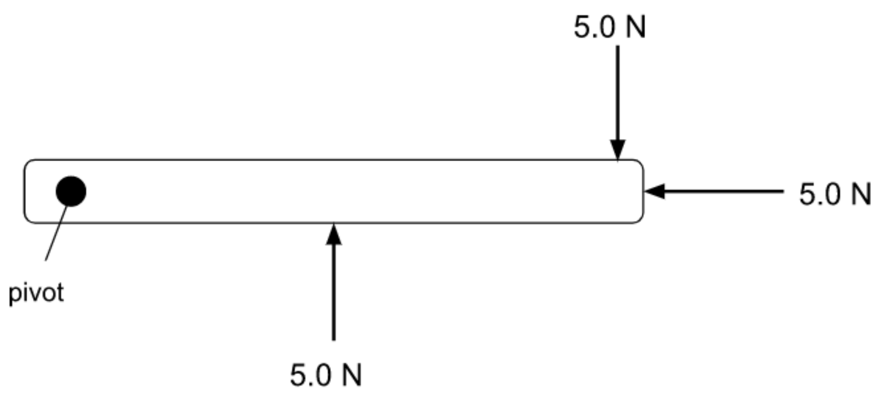
B. Collision time increases, which reduces the rate of change of momentum.

C. Collision time decreases, which increases the rate of change of momentum.

D. Collision time decreases, which reduces the rate of change of momentum .

(1 mark)

7 The diagram shows a door handle with three forces acting on it.



How will the door handle move?

A. It will turn clockwise.

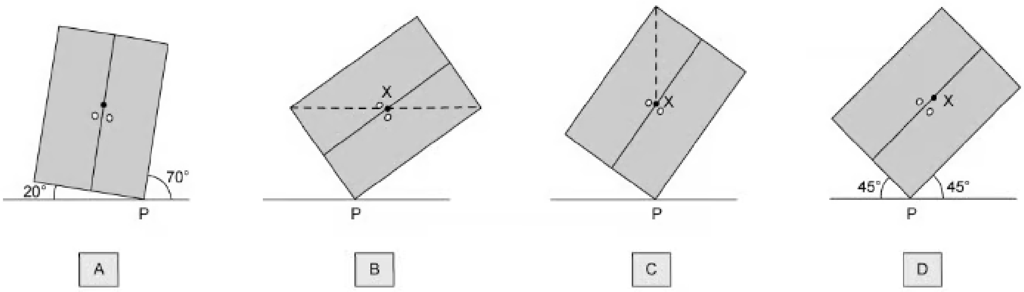
B. It will turn anticlockwise.

C. It will move to the left.

D. It will not move at all.

(1 mark)

8 A wardrobe is shown tilted at a variety of angles.



In which position is the wardrobe in equilibrium?

(1 mark)

9 A rollercoaster moving along a track has a large momentum.

A second rollercoaster has half the mass of the first rollercoaster and travels at four times the speed.

What is the momentum of the second rollercoaster compared to the first?

A. The second rollercoaster has a momentum of zero

B. The second rollercoaster's momentum is half that of the first rollercoaster's momentum

C. The second rollercoaster's momentum is twice that of the first rollercoaster's momentum

D. The second rollercoaster's momentum is four times that of the first rollercoaster's momentum

(1 mark)

10 A large truck of mass 5000 kg is travelling at 5.0 m/s. A motorbike has a mass of 200 kg. Both the truck and the motorbike have the same kinetic energy.

What is the speed of the motorbike?

A. 25 m/s

B. 10 m/s

C. 125 m/s

D. 2.5 m/s

(1 mark)

11 Which of the following energy transfers is most efficient?

A. Useful energy out = 70%

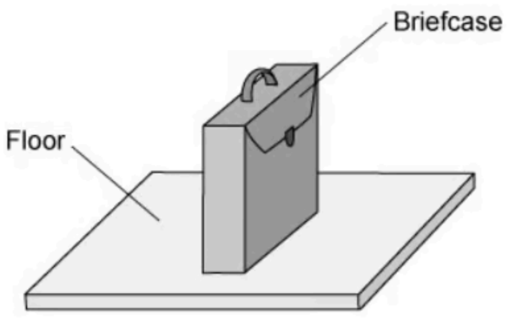
B. Wasted energy out = 45%

C. Useful energy out = 35%

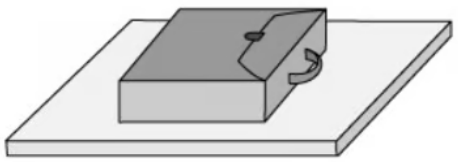
D. Wasted energy out = 25%

(1 mark)

12 A briefcase, with flat, rectangular sides rests on the floor as shown in the diagram.



The briefcase is now turned so that it rests with its large, flat side on the floor.



How has the change affected the force on the floor, and the pressure exerted by the briefcase on the floor?

	Force	**Pressure**
A	unchanged	decreased
B	unchanged	unchanged
C	decreased	decreased
D	decreased	unchanged

(1 mark)

13 A beaker of water, at its boiling point of 100 ^{o}C, is heated so that the liquid starts to become a gas, also at 100 ^{o}C.

Which row of the table correctly compares the liquid phase with the gas phase under these conditions?

	Average distance between the particles	**Average speed of the particles**
A	the same in the liquid and the gas	the same in the liquid and the gas
B	greater in the liquid	greater in the gas
C	greater in the gas	greater in the liquid
D	greater in the gas	the same in the liquid and the gas

(1 mark)

14 A car tyre is inflated with air.

As the car drives, the temperature of the tyre increases, which also causes the temperature of the air inside the tyre to increase. This increases the pressure of the air inside the tyre.

Which of the following is a correct explanation as to why?

A. The mass of air inside the tyre has increased.

B. The area of the tyre has decreased, thus increasing the pressure.

C. The air molecules collide with each other less frequently.

D. The average kinetic energy of the air molecules has increased.

(1 mark)

15 Which of the following statements defines the **specific heat capacity** of a substance?

A. The energy required to raise the temperature of 1 kg of a substance by 1℃.

B. The energy per ℃ required to raise the temperature of a substance.

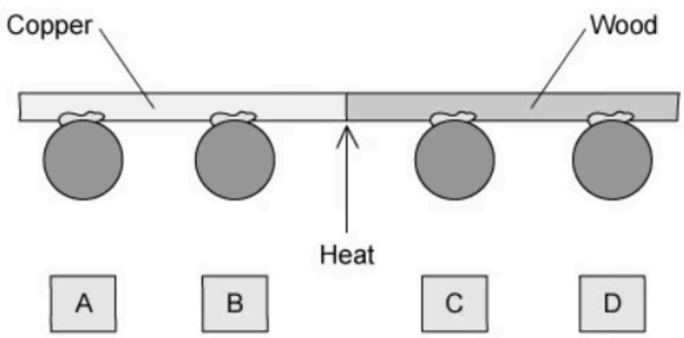
C. The increase in length of 1kg of a substance when its temperature is increased by 1℃.

D. The energy required to change 1kg of a substance from a liquid to a gas.

(1 mark)

16 In a very common classroom Physics demonstration, a teacher sets up a rod, as shown in the diagram. Half of the rod is copper, the other half is wood.

She sticks four lead balls to the rod using wax. The wax melts when its temperature rises a small amount, causing the ball to fall off.



The teacher heats the rod in the centre, with something more gentle than a Bunsen burner, that will not burn the wood.

Which ball falls off first?

(1 mark)

17 In a children's experiment, a saucer of water is left outside to evaporate.

Which row shows weather conditions that will each cause the rate of evaporation from the saucer to increase?

	Temperature	Wind speed
A	warm	windy
B	warm	still
C	cool	windy
D	cool	still

(1 mark)

18 Which of the following statements about energy transfer is correct?

A. Metals conduct thermal energy well because their electrons are not free to move.

B. Infrared radiation cannot travel through a vacuum.

C. Warm fluids rise because their particles move further apart.

D. Convection only occurs in gases.

(1 mark)

19 The speed of a wave is doubled. The frequency remains the same.

What happens to the wavelength of the wave?

A. It doubles

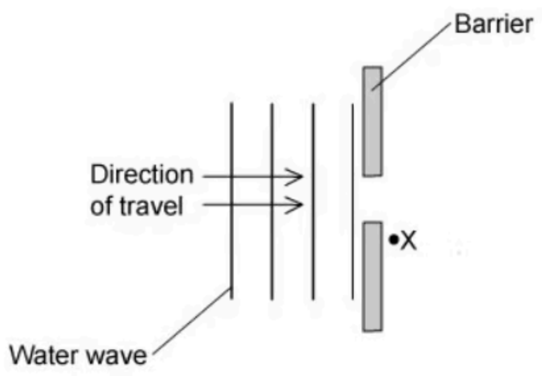
B. It halves

C. It remains the same

D. It becomes four times as large

(1 mark)

20 What is the name given to the effect that allows the wavefronts to reach point X, shown on the diagram?



A. Reflection

B. Refraction

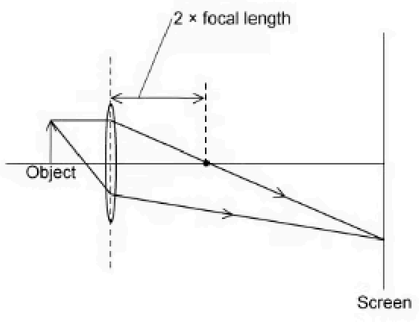
C. Diffraction

D. Dispersion

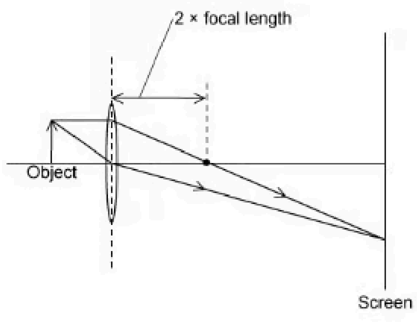
(1 mark)

21 Which diagram correctly shows the formation of a real image on a screen?

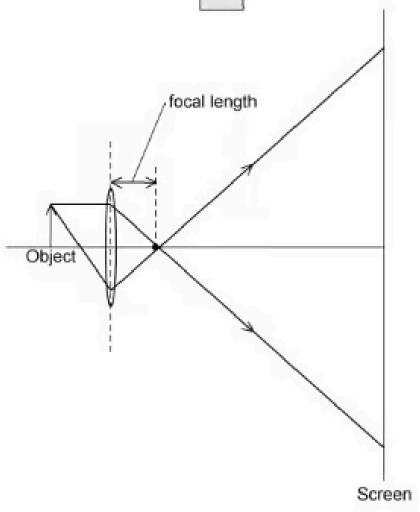
A



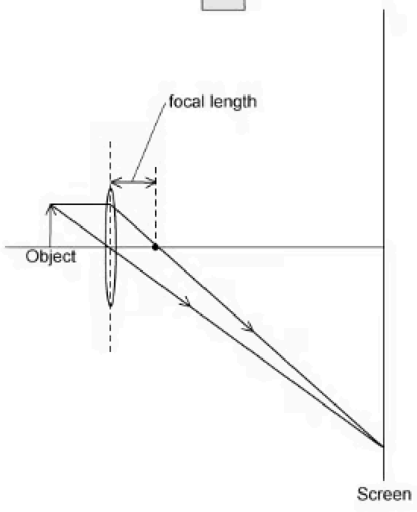
B



C

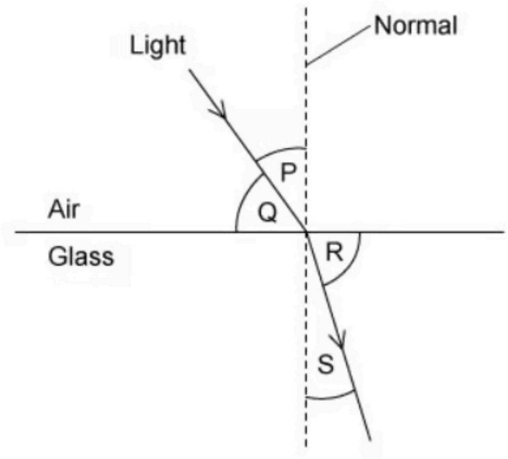


D



(1 mark)

22 The diagram shows light passing from air into glass. A number of angles have been labelled on the diagram.



Which of the equations could be used to correctly calculate the refractive index, *n*, of the glass?

A. $\frac{\sin P}{\sin S}$

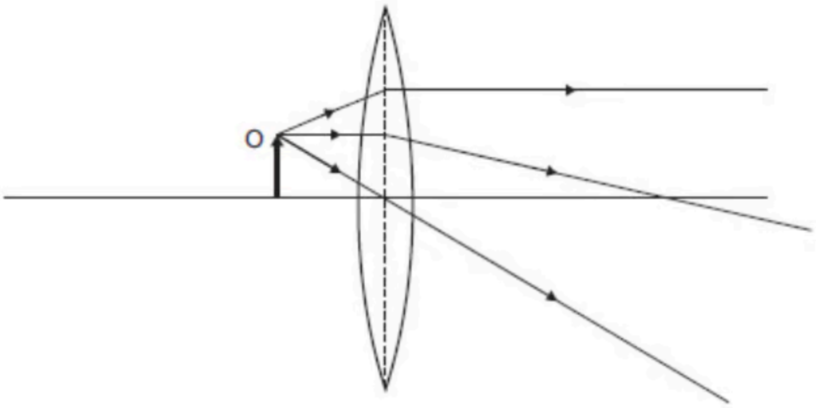
B. $\frac{\sin Q}{\sin R}$

C. $\frac{\sin P}{\sin R}$

D. $\frac{\sin Q}{\sin S}$

(1 mark)

23 The diagram shows an object positioned close to a thin converging lens (inside the focal distance). Three light rays are shown leaving the object and passing through the lens.



Which row in the table correctly describes the properties of the image?

	location	size	real / virtual
A	to the left of O	reduced	real
B	to the right of O	enlarged	virtual
C	to the left of O	enlarged	virtual
D	at O	the same as O	real

(1 mark)

24 Which of the following types of electromagnetic radiation has the highest speed in a vacuum?

A. Microwaves

B. Radio waves

C. Visible light

D. None of the above

(1 mark)

25 Ultrasound is high-pitched and cannot usually be heard by human ears. However, ultrasound does have many uses.

Which statement, or statements, in the list below are correct uses of ultrasound?

1. Testing materials to find breaks or other damage
2. Medical scanning of organs and soft tissues
3. Radar systems to find distance and velocity

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

(1 mark)

26 Which of the following is **not** a use of ultrasound?

A. Scanning a fetus during pregnancy

B. Non-destructive testing of materials such as to find leaks in copper pipes

C. Mapping with sonar

D. Identifying broken bones

(1 mark)

27 Which statement about a permanent bar magnet is **incorrect**?

A. The S-pole attracts the N-pole of another magnet

B. The S-pole repels the S-pole of another magnet

C. It cannot repel a non-magnetic material

D. It is made from a soft magnetic material (such as soft iron)

(1 mark)

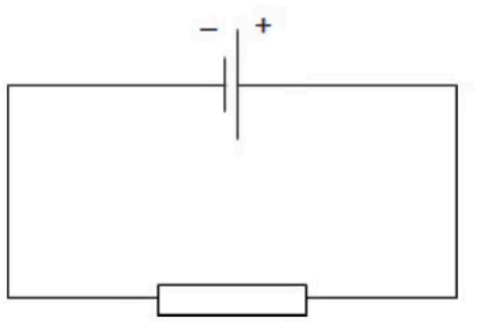
28 A PVC (plastic) rod is rubbed with a nylon cloth. This process causes electrons to be transferred between the rod and the cloth, causing both objects to become charged.

Which row gives the correct nature of the charges on both the cloth and the rod, and the effect the objects have on each other after becoming charged?

	Charges on rod and cloth	**Effect**
A	the same	repel
B	the same	attract
C	opposite	repel
D	opposite	attract

(1 mark)

29 A student sets up a circuit as shown in the diagram.



A charge of 4.9 C flows through the resistor in 0.7 s.

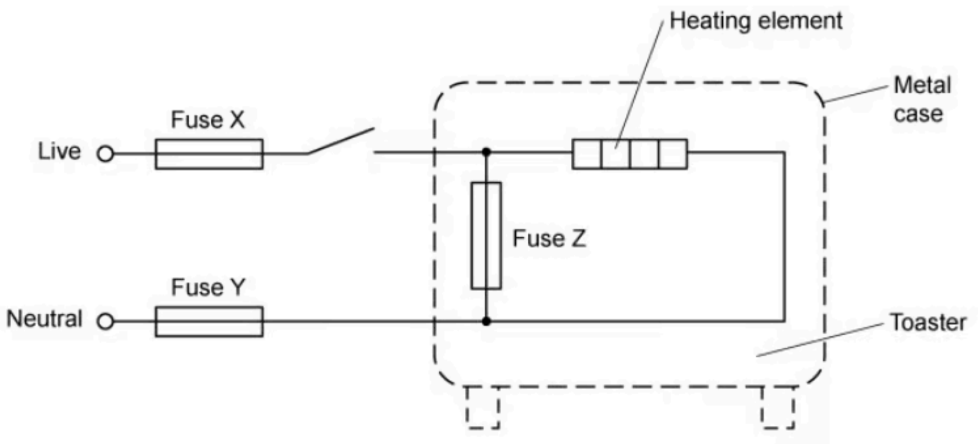
Which row shows the correct current, direction of electron flow and direction of conventional current through the resistor?

	current / A	direction of electron flow	direction of conventional current
A	7.00	Left to right	Right to left
B	3.43	Left to right	Right to left
C	7.00	Right to left	Right to left
D	3.43	Right to left	Right to left

(1 mark)

30 The diagram shows a toaster that has been wired up.

Three fuses have been connected to the toaster, but only one or two have been connected correctly.



Which of the fuses has been positioned correctly?

A. Fuse X and fuse Y

B. Fuse X only

C. Fuse Z only

D. Fuse X and fuse Z

(1 mark)

31 The electromotive force (e.m.f.) of a mobile phone battery is 3.7 V.

What does this mean?

A. 3.7 J is the maximum energy the battery can provide in 1.0 s.

B. 3.7 J is the total energy the battery can provide before it has to be recharged.

C. 3.7 J of energy is provided by the battery to drive a charge of 1.0 C around a complete circuit.

D. 3.7 J of energy is provided by the battery to drive a current of 1.0 A around a complete circuit.

(1 mark)

32 An oven is wired into somebody's kitchen. The oven is protected by a fuse, and functions normally, but the owner notices that the wires that connect the oven to the mains electricity supply get quite hot.

What could be done to reduce the temperature of the wires?

A. Add a circuit breaker to the circuit.

B. Put thicker insulation on the wires.

C. Replace the wires with thicker diameter wires.

D. Replace the wires with thinner diameter wires.

(1 mark)

33 When a wire is moved through a magnetic field, an e.m.f. is induced in the wire.

Which of the following devices uses this effect?

A. electric motor

B. electricity generator

C. photon accelerator

D. transformer

(1 mark)

34 After electricity is generated at a power station, the voltage is increased significantly for transmission across the country via the national grid.

What is the advantage of transmitting electricity at very high voltages?

A. It makes the electricity flow more quickly.

B. It increases the efficiency of the electricity transfer.

C. It produces more power.

D. It is safer to transmit electricity at high voltage.

(1 mark)

35 Which device uses a split ring commutator?

A. d.c. motor

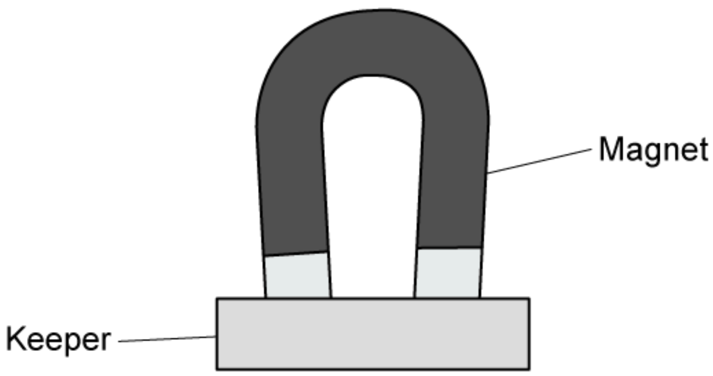
B. transformer

C. relay

D. a.c. generator

(1 mark)

36 The diagram shows two bar magnets, stored with metal keepers across the ends. The keepers help to keep the magnets magnetised.



The material used for the keepers becomes strongly magnetised when placed in contact with the magnets, but does not remain magnetised when taken away from the magnets.

What is a suitable metal to use for the magnets and what is a suitable metal to use for the keepers?

	Metal for keeper	Metal for magnet
A	steel	steel
B	iron	steel
C	steel	iron
D	iron	iron

(1 mark)

37 A nucleus of uranium ($^{238}_{92}U$) is unstable and decays by emitting an **α**-particle.

Which equation correctly describes this process?

A. $^{238}_{92}U \rightarrow {}^{4}_{2}\alpha + {}^{234}_{90}Th$

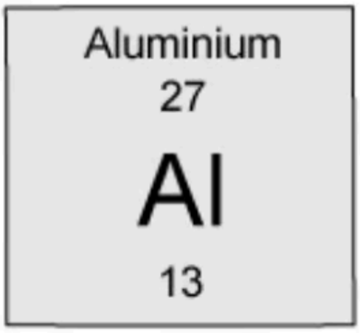
B. $^{238}_{92}U \rightarrow {}^{4}_{2}\alpha + {}^{234}_{90}U$

C. $^{238}_{92}U + {}^{4}_{2}\alpha \rightarrow {}^{234}_{90}Th$

D. $^{238}_{92}U \rightarrow {}^{4}_{2}\alpha + {}^{234}_{88}Ra$

(1 mark)

38 How many neutrons are in a nucleus of aluminium? The chemical symbol is given below:



A. 13

B. 27

C. 14

D. 40

(1 mark)

39 Comets travel faster within the Solar System than they do when they are outside it.

Which of the following gives the correct reason for this?

A. A comet near to the Sun has more gravitational potential energy.

B. Comets closer to the Sun transfer gravitational potential energy to kinetic energy due to conservation of energy.

C. A comet which is outside the Solar System has less energy than one which is passing through it.

D. Comets closer to the Sun transfer kinetic energy to gravitational potential energy due to conservation of energy.

(1 mark)

40 What is the process which makes stable stars such as the Sun shine, and which element produced in the reaction?

	Process	Element
A	fusion	oxygen
B	fission	hydrogen
C	fusion	helium
D	fission	hydrogen

(1 mark)